

B.Sc. 1st Semester (Honours) Examination-2022-23

COMPUTER SCIENCE

Course ID : 11512 Course Code : SH/CSC/102/C-2

Course Title :

**Computer System Organization and Architecture (New)
Computer System Architecture (Old)**

Time : 1 Hour 15 Minutes

Full Marks : 25

The figures in the right hand margin indicate full marks.

*Candidates are required to give their answers in their
own words as far as practicable*

Unit-I

1. Answer any five questions :

1×5=5

~~(a)~~ Write down the truth table of XOR gate.

~~(b)~~ Define Combinational circuit.

~~(c)~~ Represent the decimal number 21.27 to binary.

~~(d)~~ Why is cache memory used?

~~(e)~~ What is decoder?

1.0 + 0.1

0 + 0

0

(Turn Over)

- (f) Define bus.
- (g) Define stack.
- (h) Write down two applications of associative memory.

Unit-II

2. Answer **any two** questions : 5×2=10

- (a) Describe direct mapping process in Computer Architecture?
- (b) Design D flip-flop.
- (c) Design a magnitude comparator.
- (d) Write short note on different types of Address Bus.

10101
11011

2127 Unit-III

3. Answer **any one** question : 10×1=10

- (a) Describe RISC and CISC architectures of computer. 5+5
- (b) Implement a full-subtractor circuit. Obtain the simplified expression of the function. 5+5

$$F(w,x,y,z) = \sum (4,6,7,8,9,10,11,14)$$

BSC 1st semester examination -2022

Subject: Computer Science Course Title: : Computer Architecture

Full Marks: 10

Time: 30 MIN

Answer any five questions:

- ~~1.~~ State Demorgan's law.
- ~~2.~~ Convert the following question Hexadecimal to Octal number.
i) $(2CD)_{16}$ ii) $(3EC)_{16}$
- ~~3.~~ Write down the truth table of X-OR gate.
- ~~4.~~ Why NAND gate is called universal gate?
- ~~5.~~ 2's Complement the $(11011101)_2$
- ~~6.~~ Convert the following question i) $(743)_8 = (?)_2$ ii) $(ABC)_{16} = (?)_2$
- ~~7.~~ Prove the Distributive law.
- ~~8.~~ Implement the expression $(AB+B'C+CD')$ using basic gate.

B.Sc. 1st Semester (Honours) Examination-2022-23

COMPUTER SCIENCE (PRACTICAL)

Course ID : 11522 Course Code : SH/CSC/102/C-2P

Course Title :

**Computer System Organization and Architecture
(New)**

**Computer System Architecture
(Old)**

Time : 2 Hours

Full Marks : 15

The figures in the right hand margin indicate full marks.

*Candidates are required to give their answers in their
own words as far as practicable.*

Problem-10, Viva + LNB-05

1. Answer any one question :

10×1=10

(a) Implement XOR gate using NAND gates.

(Turn Over)

2

		AND	NAND
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

(b) Implement a 4-bit Binary Adder using IC 7483.

(c) Implement 8×1 Mux using IC 74153.

(d) Implement Full-Subtractor using NAND gates.

(e) Implement J-K Flip-Flop using NAND gates.

(f) Implement a Half-Adder using NAND gates.

(g) Implement S-R Flip-Flop using NAND gates.

(h) Implement a 2 bit Comparator using Basic gates.

(i) Implement Full Adder using NAND gates.

(j) Implement the function $F = ABC + DEF$ using IC 7411.